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A Report On

"Academic Performance Analysis Dashboard (APAD)"

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INTRODUCTION

The Academic Performance Analysis Dashboard (APAD) is a web-based application developed to provide a secure, efficient, and user-friendly platform for tracking and analysing student academic performance. The system is designed using HTML, CSS, and JavaScript for the frontend, PHP for server-side scripting, and MySQL for database management. It aims to replace traditional manual record-keeping processes with an automated solution that ensures accuracy, transparency, and time efficiency.

The application allows users (faculty or administrators) to register and log in securely using authenticated credentials. Password security is maintained through proper validation and encryption techniques. Once authenticated, users can add students, record marks across subjects, and generate performance reports. The system ensures that data remains consistent and that each user can only access records they are authorised to manage, maintaining the integrity of academic data.

An administrative panel is included to manage users, students, and performance records. The admin has the authority to view, edit, or delete any record and to toggle user roles between *admin* and *user*. The system also provides features for data display, search, filtering, and automated report generation with grading, enabling efficient tracking and analysis of student results.

The project implements core functionalities such as CRUD operations, form validation, session management, and database integration with relational tables. A responsive dark-mode user interface ensures accessibility across different devices. Overall, APAD demonstrates the practical implementation of web development concepts to build a reliable and scalable academic performance tracking platform.

PROBLEM STATEMENT

Traditional academic record systems, whether paper-based or simple spreadsheet methods, often suffer from several critical issues such as time consumption, lack of transparency, risk of human error, and difficulty in managing and aggregating performance data efficiently. Manual processes require significant administrative effort, are prone to inconsistencies, and do not provide real-time access to results or comparative analysis.

In many educational institutions, there is a lack of a secure and reliable digital platform to track student performance over multiple subjects and semesters. Existing solutions may not enforce proper user authentication, allowing unauthorised access or unintended modification of academic records, which compromises the fairness and accuracy of the evaluation process.

Additionally, managing student profiles, subject-wise marks, and performance summaries manually or through unstructured systems leads to data inconsistency and inefficiency. There is also a need for an organised interface that allows administrators to oversee the entire system while providing regular users with a simple way to manage their own student data.

Therefore, there is a need to develop a secure, efficient, and user-friendly Academic Performance Analysis Dashboard that ensures proper authentication, role-based access control, data integrity, automated grading, and real-time access to performance reports through a centralised database-driven application.

SYSTEM ARCHITECTURE

The Academic Performance Analysis Dashboard follows a three-tier architecture consisting of the Presentation Layer (Client Side), Application Layer (Server Side), and Database Layer. This architecture ensures separation of concerns, better scalability, and maintainability of the system.

1. Presentation Layer (Client Side):

This layer is responsible for user interaction and interface. It is developed using HTML5, CSS3 (custom dark-mode design system), and Vanilla JavaScript. Users access the system through a web browser where they can register, log in, add students, record marks, and view performance dashboards. The interface provides forms for input and displays data such as student lists, records, and chart-based summaries using Chart.js v4.

2. Application Layer (Server Side):

The application layer is implemented using PHP 8+ and handles all business logic of the system. It processes user requests received from the client side, performs validation, manages user sessions, and controls access based on roles (admin or user). This layer ensures that only authenticated users can access data and that regular users can only manage their own enrolled students. It also manages CRUD operations for users, students, records, and reports.

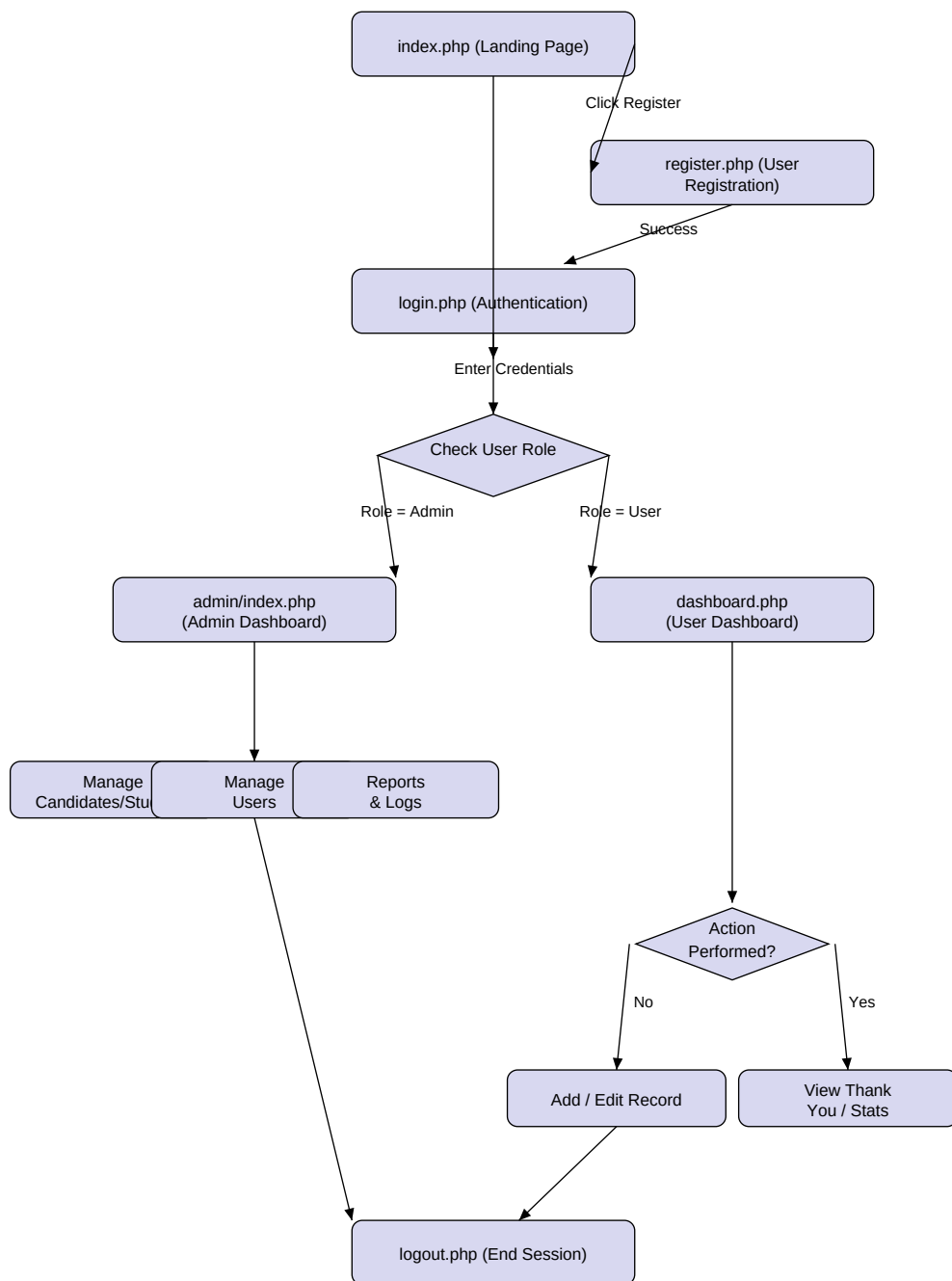
3. Database Layer:

The database layer uses MySQL 5.7+ / MariaDB 10+ to store and manage all system data. It includes four main tables: Users, Students, Records, and Reports. The database maintains relationships between these tables using primary and foreign keys, enforcing referential integrity. It supports efficient retrieval of information for reporting, filtering, and chart generation purposes.

Working Process:

When a user interacts with the system, the request is sent from the browser to the PHP server. The server processes the request, communicates with the MySQL database to store or retrieve data, and sends the response back to the user interface. For example, when generating a report, the system calculates the average percentage, assigns a grade, and stores the report in the database for future reference. This structured architecture improves system security, reliability, and performance while making the application easier to manage and extend in the future.

FLOW CHART



DATABASE DESIGN

Tables used in Academic Performance Analysis Dashboard

1. Users
2. Students
3. Records
4. Reports

USERS		
int	id	PK
string	name	
string	email	
string	password	
string	role	admin/user
timestamp	created_at	

STUDENTS		
int	id	PK
string	student_code	
string	name	
string	department	
int	semester	
int	enrolled_by	FK→users

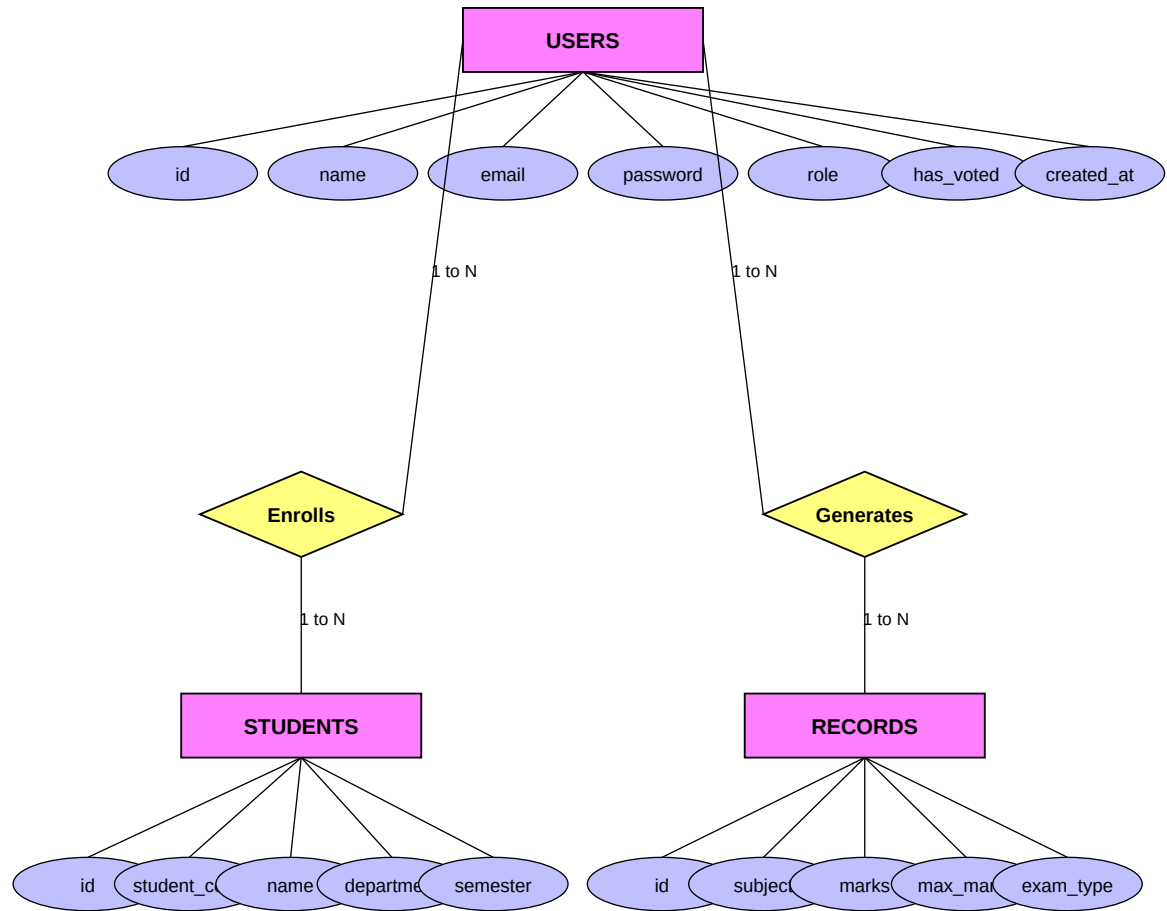
enrolled_by →

student_id →

RECORDS		
int	id	PK
int	student_id	FK→students
string	subject	
int	marks	
int	max_marks	
string	exam_type	
date	exam_date	

REPORTS		
int	id	PK
int	student_id	FK→students
float	avg_percentage	
string	grade	
string	performance_status	
int	generated_by	FK→users

E R DIAGRAM



FEATURES & SNAPSHOTS

The following section highlights the key features implemented in the Academic Performance Analysis Dashboard and describes the interface snapshots of each major module.

Authentication Module

The system provides a secure login and registration page. New users can register with name, email, and password. A real-time password strength meter provides visual feedback. Upon login, the server validates credentials against bcrypt-hashed passwords and establishes a PHP session for the user.

Dashboard

The main dashboard displays at-a-glance statistics via stat cards: total students, total records, total reports, and average performance. Two Chart.js visualisations — a bar chart (marks by subject) and a doughnut chart (grade distribution) — provide an immediate overview of academic trends.

Student Management

The Students module offers full CRUD functionality. Faculty can add students with a unique student code, name, department, and semester. The listing page supports multi-field search and filtering. Each student entry shows quick action buttons for view, edit, and delete.

Performance Records

The Records module allows entry of subject-wise marks for any enrolled student. Fields include subject name, marks obtained, maximum marks, exam type (Internal / Mid-Term / Final), exam date, and optional remarks. A live preview card calculates the percentage and grade in real time as data is entered.

Report Generation

The Reports module auto-generates performance summaries for selected students. The system calculates the average percentage across all recorded subjects, assigns a grade (A+ to F) according to the configured grade scale, and labels the performance status (Excellent / Good / Average / Poor / Fail). Reports are stored in the database and can be viewed or deleted.

Admin Panel

Administrators have access to two additional panels: All Records (a system-wide view of every performance record regardless of creator) and User Management (ability to view all registered users, toggle their role between admin and user, or delete accounts).

Grade Scale

The system uses the following grade scale to evaluate student performance:

Grade	Percentage
A+	≥ 90%
A	75 – 89%
B	60 – 74%
C	45 – 59%

D	33 – 44%
F	< 33%

CONCLUSION

The Academic Performance Analysis Dashboard (APAD) provides a secure and efficient platform for tracking and evaluating student academic performance. It ensures that only authenticated users can access and manage data, with role-based access control maintaining the separation between administrative and regular-user capabilities. The system successfully implements database management, CRUD operations, user authentication, and automated report generation with grading using PHP and MySQL. A responsive dark-mode interface powered by Chart.js enhances usability across devices. Overall, the project demonstrates a practical application of web development concepts and can be further improved with advanced features such as email notifications, graphical trend analysis over multiple semesters, and integration with institutional portals.

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